

# Experimental Evaluation of Low-Resource Retrieval Architectures for Scientific Literature

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**Abstract**—This paper presents a systematic experimental evaluation of retrieval architectures for scientific literature under strict low-resource constraints — CPU-only execution and free-tier cloud compute. We compare two families of sentence embedding models (`all-MiniLM-L6-v2` and `bge-small-en-v1.5`) across three chunking strategies and measure Recall@3, Recall@5, Precision@5, and Mean Reciprocal Rank (MRR) on a query set stratified by difficulty. We further evaluate a lightweight cross-encoder reranker and introduce the Adaptive Retrieval Score (ARS), a hybrid ranking mechanism that compensates for known limitations of dense semantic retrieval through lexical, technical, and structural signals. Results show that retrieval-specialized embeddings (BGE-small) consistently outperform general-purpose embeddings by up to 53% in MRR on hard queries without reranking, while cross-encoder reranking elevates all configurations above 0.89 MRR and reduces the maximum inter-configuration spread from 0.294 to 0.035. The reranker introduces approximately 300 ms of latency overhead per query, representing the primary deployment trade-off. These findings provide actionable guidance for building scientific retrieval systems under academic resource constraints.

**Index Terms**—retrieval-augmented generation, semantic search, sentence embeddings, cross-encoder reranking, low-resource NLP, scientific literature, chunking strategies

## I. INTRODUCTION

Retrieval-Augmented Generation (RAG) systems have become a standard approach for grounding language model outputs in domain-specific knowledge. However, most published evaluations assume access to GPU infrastructure, large embedding models, and extensive annotated corpora. These assumptions are impractical in academic settings with limited computational budgets — such as the Universidad de Antioquia — where research must execute under CPU-only or free-tier cloud constraints.

Scientific literature retrieval presents challenges beyond standard question-answering benchmarks. Papers contain dense mathematical notation, domain-specific acronyms, and technically precise vocabulary where semantically similar passages may be factually incorrect. A system that retrieves the nearest embedding vector may still return a passage that shares terminology but describes a different concept.

This work addresses four research questions: (1) How does embedding specialization — general-purpose vs. retrieval-trained — affect retrieval quality on scientific text? (2) How do chunking strategies interact with embedding choice? (3) Can a

lightweight cross-encoder reranker close the performance gap between weaker and stronger first-stage retrievers? (4) What is the latency cost of reranking in CPU-only environments?

The contributions of this work are: (i) an experimental comparison of MiniLM and BGE embedding families on a scientific literature corpus; (ii) a difficulty-stratified evaluation protocol (easy, medium, hard); (iii) a latency benchmark across all six configurations; and (iv) the Adaptive Retrieval Score (ARS), a theoretically motivated hybrid ranking mechanism addressing the limitations of pure dense retrieval.

## II. RELATED WORK

Dense passage retrieval [6] established bi-encoder architectures as the dominant paradigm for scalable semantic search. Sentence-BERT [2] introduced siamese fine-tuning of BERT to produce sentence embeddings suitable for semantic similarity tasks. MiniLM [3] compressed large transformer models through deep self-attention distillation, transferring query-key attention distributions and value-relation representations from a teacher into a compact student model. The BGE embedding family [4] extended this line by applying contrastive training with hard negatives specifically optimized for retrieval discrimination.

Cross-encoder rerankers [5] have been shown to substantially improve first-stage retrieval by jointly encoding query and document, enabling token-level attention interactions unavailable in bi-encoders. Systematic evaluations of RAG chunking strategies remain limited; most RAG literature selects fixed-size chunking without ablation. The original Transformer architecture [1] underpins all models evaluated in this work.

Our work differs in focusing on the low-resource constraint, evaluating scientific text with difficulty-stratified queries, and providing latency measurements alongside retrieval quality metrics.

## III. ARCHITECTURAL MOTIVATION AND MODEL SELECTION

This project was designed to understand the architectural behavior and representational limitations of embedding models before selecting experimental components. Foundational literature on Transformer architectures [1], sentence embeddings [2], MiniLM distillation [3], and retrieval-specialized

encoders [4] was studied to ground every design decision. This theoretical study directly informed both model selection and the proposal of ARS.

#### A. Why *all-MiniLM-L6-v2*?

*all-MiniLM-L6-v2* was selected as a computationally efficient baseline. The MiniLM architecture uses Deep Self-Attention Distillation to compress large transformer language models into smaller encoders, transferring query-key attention distributions and value-relation representations from teacher to student rather than simply distilling output predictions. Following compression, the model is further optimized within the Sentence-Transformers framework to generate sentence-level embeddings. This dual-stage process makes MiniLM a strong low-resource semantic baseline, with embedding times of approximately 13–14 ms per query on CPU.

#### B. Why *bge-small-en-v1.5*?

Unlike general-purpose sentence embeddings, BGE is optimized for retrieval through contrastive training with hard negatives — semantically similar but factually distinct examples that force the model to learn finer-grained discrimination. This property is particularly valuable in scientific literature, where passages frequently share surface vocabulary while describing different methods or results. BGE embeddings require approximately 22 ms per query on CPU, a modest overhead relative to MiniLM.

The methodological contrast is deliberate: general-purpose lightweight semantic embeddings vs. retrieval-specialized semantic embeddings under identical low-resource constraints.

#### C. Why Not Larger Models?

Larger models such as BGE-M3 were excluded because they introduce latency and memory requirements incompatible with the primary experimental constraint — CPU-only and Colab Free execution. Methodological comparability requires that hardware limits remain constant across all configurations.

### IV. METHODOLOGY

#### A. Corpus and Chunking Strategies

The corpus consists of foundational deep learning papers (attention mechanisms, CNN, etc.) processed through three chunking strategies:

- **Fixed-small:** chunks of  $\sim 256$  tokens with overlap.
- **Fixed-large:** chunks of  $\sim 512$  tokens with overlap.
- **Semantic:** boundary detection based on sentence-level cosine similarity shifts.

Each strategy produces an independent FAISS vector index used for approximate nearest-neighbor search.

#### B. Evaluation Protocol

A query set of 30 questions was designed and manually stratified into three difficulty levels:

- **Easy:** unambiguous terminology with direct correspondence in the corpus.
- **Medium:** require synthesis across multiple passages.

TABLE I  
RETRIEVAL METRICS WITHOUT RERANKING

| Configuration      | R@3          | R@5          | P@5   | MRR          |
|--------------------|--------------|--------------|-------|--------------|
| minilm-small-fixed | 0.733        | 0.833        | 0.433 | 0.583        |
| minilm-large-fixed | 0.800        | 0.833        | 0.567 | 0.736        |
| minilm-semantic    | 0.700        | 0.700        | 0.340 | 0.556        |
| bge-small-fixed    | 0.900        | 0.933        | 0.667 | 0.831        |
| bge-large-fixed    | 0.867        | 0.900        | 0.693 | 0.801        |
| bge-semantic       | <b>0.900</b> | <b>0.900</b> | 0.607 | <b>0.850</b> |

- **Hard:** require precise lexical or mathematical matching.

Metrics reported are Recall@3 (R@3), Recall@5 (R@5), Precision@5 (P@5), and Mean Reciprocal Rank (MRR). Difficulty-stratified MRR is also reported separately.

#### C. Adaptive Retrieval Score (ARS)

Dense bi-encoders capture semantic relatedness but may fail under queries requiring precise lexical or mathematical matching — a limitation arising directly from the self-attention training objective. This observation motivated the design of ARS, a post-retrieval hybrid scoring mechanism applied to the top- $k$  candidates:

$$\text{ARS}(q, d) = \alpha \cdot \text{sem}(q, d) + \beta \cdot \text{lex}(q, d) + \gamma \cdot \text{tech}(d) \quad (1)$$

where *sem* is cosine semantic similarity, *lex* is token overlap precision, *tech* is a technical density indicator (density of domain-specific terms), and *coh* is a structural coherence score based on chunk boundary quality. ARS is not a replacement for dense retrieval but a compensatory re-scorer that addresses its architectural limitations.

#### D. Cross-Encoder Reranking

Bi-encoders encode query and document independently, enabling scalable retrieval through cosine similarity but precluding direct query-document interaction during encoding. A lightweight fine-tuning cross-encoder over scientific documents (*ms-marco-MiniLM-L-6-v2*) is applied to the top-10 retrieved candidates, jointly processing [Query; Document] through the full attention mechanism. This architectural difference enables richer contextual reasoning at the cost of linear latency scaling with candidate count.

### V. RESULTS

#### A. Retrieval Quality Without Reranking

Table I presents retrieval metrics for all six configurations without reranking.

BGE configurations outperform MiniLM across all metrics. The best first-stage configuration (*bge-semantic*, MRR = 0.850) exceeds the best MiniLM configuration (*minilm-large-fixed*, MRR = 0.736) by 15.5%. On hard queries, this gap widens substantially: *bge-small-fixed* achieves MRR = 0.656 vs. 0.438 for *minilm-large-fixed* — a 50% relative improvement. Semantic chunking benefits BGE but hurts MiniLM,

TABLE II  
DIFFICULTY-STRATIFIED MRR WITHOUT RERANKING

| Configuration      | Easy         | Medium       | Hard         |
|--------------------|--------------|--------------|--------------|
| minilm-small-fixed | 0.650        | 0.642        | 0.410        |
| minilm-large-fixed | 0.900        | 0.799        | 0.438        |
| minilm-semantic    | 0.600        | 0.736        | 0.229        |
| bge-small-fixed    | 1.000        | 0.806        | 0.656        |
| bge-large-fixed    | 1.000        | 0.833        | 0.504        |
| bge-semantic       | <b>0.950</b> | <b>0.917</b> | <b>0.625</b> |

TABLE III  
RETRIEVAL METRICS WITH CROSS-ENCODER RERANKING

| Configuration      | R@3          | R@5          | P@5          | MRR          |
|--------------------|--------------|--------------|--------------|--------------|
| minilm-small-fixed | 0.967        | 0.967        | 0.860        | 0.922        |
| minilm-large-fixed | 0.933        | 0.933        | 0.827        | 0.894        |
| minilm-semantic    | 0.900        | 0.933        | 0.813        | 0.907        |
| bge-small-fixed    | <b>0.967</b> | <b>1.000</b> | <b>0.853</b> | <b>0.929</b> |
| bge-large-fixed    | 0.933        | 0.933        | 0.827        | 0.911        |
| bge-semantic       | 0.933        | 0.967        | 0.840        | 0.923        |

suggesting that chunking strategy interacts with the embedding training objective.

#### B. Difficulty-Stratified Analysis

Table II presents MRR disaggregated by query difficulty.

All configurations approach ceiling performance on easy queries. Hard queries reveal the largest gaps, confirming that retrieval-specialized embeddings provide meaningful advantages precisely when lexical precision is required — the scenario that motivated ARS design.

#### C. Retrieval Quality With Reranking

Table III presents retrieval metrics after cross-encoder reranking.

Reranking produces substantial gains across all configurations. MRR improvements range from +0.073 (bge-semantic: 0.850  $\rightarrow$  0.923) to +0.339 (minilm-small-fixed: 0.583  $\rightarrow$  0.922). Crucially, the maximum MRR spread across configurations drops from 0.294 (without reranking) to 0.035 (with reranking), demonstrating that the cross-encoder compensates effectively for first-stage retrieval weaknesses when relevant documents appear in the top-10 candidate set.

#### D. Latency Analysis

Table IV presents embedding and FAISS search latency. All BGE configurations embed queries in 21–23 ms; MiniLM is approximately 40% faster at 13–14 ms. FAISS search adds under 0.2 ms across all configurations.

Cross-encoder reranking introduces 295–333 ms of overhead per query, as shown in Table V. This represents a 15–22 $\times$  increase over base retrieval latency, acceptable for interactive or offline use but requiring caching or asynchronous pipelines for sub-100 ms real-time requirements.

TABLE IV  
EMBEDDING AND SEARCH LATENCY (MS)

| Configuration      | Embed (ms)  | Search (ms)  |
|--------------------|-------------|--------------|
| minilm-small-fixed | $\sim$ 13.7 | $\sim$ 0.207 |
| minilm-large-fixed | $\sim$ 13.1 | $\sim$ 0.106 |
| minilm-semantic    | $\sim$ 13.9 | $\sim$ 0.185 |
| bge-small-fixed    | $\sim$ 22.0 | $\sim$ 0.183 |
| bge-large-fixed    | $\sim$ 22.3 | $\sim$ 0.102 |
| bge-semantic       | $\sim$ 22.3 | $\sim$ 0.172 |

TABLE V  
RERANKER LATENCY OVERHEAD (SAMPLED QUERIES)

| Difficulty | No-Rerank (ms) | Rerank (ms) | Overhead (ms) |
|------------|----------------|-------------|---------------|
| Easy       | 15.53          | 310.60      | 295.06        |
| Medium     | 19.03          | 352.27      | 333.24        |
| Hard       | 14.77          | 318.88      | 304.11        |

## VI. DISCUSSION

#### A. Embedding Specialization Matters

The consistent superiority of BGE over MiniLM without reranking confirms that training objective — contrastive retrieval training with hard negatives — is a stronger determinant of scientific retrieval quality than architectural complexity alone. `bge-small-en-v1.5`, despite having a comparable parameter count to `all-MiniLM-L6-v2`, achieves substantially higher MRR on hard queries where precise lexical discrimination is required.

#### B. Semantic Similarity Alone Is Insufficient

Hard query results expose the fundamental limitation of dense retrieval: embedding similarity captures meaning but not precision. The ARS mechanism addresses this by introducing lexical and structural compensation signals, motivated architecturally rather than by arbitrary engineering. Future work should ablate ARS components individually to quantify each signal’s contribution.

#### C. Reranking as an Equalizer

The convergence of all configurations above MRR = 0.89 after reranking has an important practical implication: when computational budget permits reranking, the choice of first-stage embedding model becomes substantially less critical. Without reranking, `bge-small-fixed` with fixed large chunking is the recommended configuration for budget-constrained deployments.

#### D. Chunking–Embedding Interaction

Semantic chunking benefits BGE but degrades MiniLM performance, suggesting that retrieval-specialized models better handle variable-length semantically coherent boundaries, while general-purpose models prefer uniform context windows. This interaction warrants further investigation and may reflect differences in how each model distributes attention over variable-length inputs.

## VII. ETHICAL CONSIDERATIONS

### A. Stakeholders and Impact of Errors

A deployed scientific retrieval system serves researchers who rely on retrieved context for literature reviews, hypothesis generation, or factual grounding. Retrieval errors — returning plausible but incorrect passages — risk propagating incorrect citations or factual claims. The difficulty-stratified evaluation was designed to expose these failure modes explicitly. Hard queries, requiring precise lexical matching, represent the highest-risk category and should be prioritized in production evaluation protocols.

### B. Justification for Deep Learning

The use of pre-trained transformer embeddings is justified because the task — semantic retrieval over heterogeneous scientific text — requires generalizable representations unavailable from keyword matching or TF-IDF. The choice of small, distilled models ensures that the computational cost of deep learning is proportionate to the retrieval gain, avoiding unnecessary resource consumption for marginal improvements.

## VIII. WORK DIVISION

This project was developed individually by Juan Pablo González Blandón, who was responsible for literature review, architecture design, corpus preparation, experimental implementation, evaluation protocol, latency benchmarking, ARS design, and report writing.

## IX. REFLECTION

The project achieved its primary objectives: a systematic, theoretically grounded comparison of retrieval architectures under low-resource constraints. The key finding — that retrieval specialization matters more than model size for scientific text — was hypothesized from the architectural study and confirmed quantitatively.

The most significant design decision was developing ARS from the observation that self-attention embeddings have structural limitations for mathematically precise queries. Future work should include individual ablation of ARS components, evaluation on a larger and more diverse scientific corpus, and investigation of whether quantized larger models (e.g., int8 BGE-M3) become viable under the low-resource constraint.

The principal learning is that retrieval architecture design requires understanding the representational geometry of embedding models, not merely benchmarking them. The theoretical study of transformer attention directly shaped the experimental design and the interpretation of results.

## X. OPEN-SOURCE REPOSITORY

The complete codebase — including all notebooks, FAISS indexes, evaluation scripts, ARS implementation, And an a app implementation with an LLM on Ollama — is publicly available at:

<https://github.com/JuanGonzalez47/low-resource-scientific-rag>

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